

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. SYSTEM INSTALLATION AND OPERATION MANUAL

Welcome to the OptiCrop Production Engine User Guide. Follow these clear operational steps to configure and execute the tool successfully:

*** Step 1: Active Development Server Initialization**

Open your development terminal interface or VS Code command terminal inside the main root folder. Type the execution prompt command: `python app.py`

Ensure the underlying server status reads: Running on <http://127.0.0.1:5000/>

*** Step 2: Accessing the User Dashboard**

Launch any standard modern web browser application configuration and navigate directly to the local link location: <http://127.0.0.1:5000/>

*** Step 3: Entering Environmental Lab Matrix Values**

Locate the clean data input boxes arranged neatly inside the interface. Input the corresponding laboratory numeric measurements for Nitrogen (N), Phosphorus (P), Potassium (K), Temperature (°C), Humidity (%), Soil pH Level, and Rainfall (mm).

*** Step 4: Activating Prediction Inference Engine**

Click the blue button block labeled 'Optimize & Recommend Crop'. The system backend evaluates variables and displays the optimal production match recommendation instantly under 2.0 seconds.